export_model.py — technical

scripts/export_model.py · 145 lines · scripts/export_model.py

Purpose

Export plus independent verification.

Step by step

Classes and functions in source order. Signatures, line ranges, raised exceptions and settings are extracted from the code; numbered steps describe what the code does.

`main`

```
def main() -> None
```

lines 62-141 · raises RuntimeError · settings CHECKPOINT_FORMAT, PRUNED_MODEL_PATH, REPLACEMENT_METHOD

Script entry point: runs this stage end to end.

1. Source: `--source` or default (trained for bias, replaced otherwise); format: `--format` or `CHECKPOINT_FORMAT`.
2. Load the source checkpoint with the configured placement; compute reference logits on 4 calibration examples.
3. Collect experiment metadata (config, statistics manifest, selection, initialisation, training report, comparison); save to `PRUNED_MODEL_PATH`; release the model.
4. Reload the export with the custom loader (strict validation).
5. Removed-weight check: delta -> the file may contain only bias keys; full -> adapter `removed_weight_violations` on stored names/shapes; all bias keys must be present.
6. Compare reloaded logits with the reference: tolerance 1e-4 (float32) or 5e-2 (16-bit); write `export_verification.json`; raise if anything failed.

Module constants

Name	Value
ATOL	{torch.float32: 0.0001, torch.float16: 0.05, torch.bfloat16: 0.05}

Errors raised

Function	Exception	Message (f-string fields in braces)
main	RuntimeError	Export check failed: leaked={leaked} biases_present={biases_present}
main	RuntimeError	Reloaded model differs from the in-memory model (max |diff| {max_diff})